spektrafilm OFX

Usage & Reference Guide

Contents

1 Introduction

spektrafilm is a photochemical emulation plugin for DaVinci Resolve. It is designed to behave less like a stack of generic grading controls and more like a compact film laboratory: camera exposure reaches a film stock, the negative is developed into density, optional lab processes alter that density, the result is printed onto a paper or print stock, and the final material is scanned or prepared for display.

This manual is written as a guided reference. It explains the analog ideas first, then shows where those ideas live in the plugin. You do not need darkroom experience to use spektrafilm, but an understanding of core colorgrading principles is greatly beneficial.

1.1 What spektrafilm Is

spektrafilm is a process model. It uses fine-tuned publicly available film and paper profile data, spectral RGB-to-raw conversion, density-domain development, grain, halation, print exposure and scanner conversion to create a color-managed film-like rendering path. Many controls operate before the final RGB image exists. For example, print exposure changes how the print material is exposed; it is not the same thing as lifting gain after a LUT.

That distinction is the reason the plugin can feel unusual at first. In most digital grading tools, you adjust an already-visible RGB image. In a film process, you often adjust a hidden intermediate: the negative, the print exposure, the paper response, or the scanner

1.2 What spektrafilm Is Not

spektrafilm is not a single-display LUT, a simple contrast curve, or a film-grain overlay. It can export color-only SDR LUTs for compatible use cases, but the live plugin includes spatial and stochastic effects that a LUT cannot contain: grain, halation, diffusion, film-plane scaling, scanner blur, scanner sharpening, and related physical effects.

It is also not intended to replace every upstream grade. While physical film development controls within the plugin allow for basic corrections, primary shot matching, balancing and exposure cleanup still belong upstream in Resolve where we can leverage the flexibility of modern colorgrading tools. spektrafilm works best when it receives a well-managed image and then acts as a creative look-builder, emulating the film, print, and scan system at the end of your chain.

1.3 Intended Resolve Workflow

Place spektrafilm near the end of the node tree, after technical input transforms and most shot-level balancing. Treat it like the print medium or film-finish node. If you want to keep grading after the film model, use `Scene Handoff` (currently still experimental); if spektrafilm should be the final display transform, use `Display Out SDR` or `Display Out HDR`.

> **Practical starting point**
>
> For a new grade, first make sure `Input Color Space` matches the data entering the node. Then leave the film and print process near defaults, balance the image upstream, choose the intended `Output Role`, and only then begin shaping stock, print, grain, halation, and scanner behavior. spektrafilm works best with wide-gamut scene-referred or log-derived input color spaces/gamuts.

1.4 Acknowledgements

spektrafilm builds on foundational open-source imaging work. The density-emulation foundation is informed by Andrea Volpato's film modeling work, and the spectral upsampling uses Johannes Hanika's "Hanatos 2025" models. This addition to the spektrafilm family adds the Resolve-facing OFX workflow, native Metal rendering, expanded print and finishing controls, and plugin-specific creative surfaces around that base.

2 Photochemical Principles

This section explains the film-lab ideas behind spektrafilm. That way you can make more informed decisions when working with spektrafilm and translate your knowledge from traditional color grading principals to the digital darkroom.

2.1 Exposure and Stops

Film records light exposure, not display brightness. A stop of light is a doubling or halving of exposure. `+1 EV` means twice as much exposure; `-1 EV` means half as much.

In spektrafilm, `Exposure EV` appears in both the Film and Print groups. They are deliberately different:

- Film `Exposure EV` acts like camera exposure before the negative is developed.
- Print `Exposure EV` acts like enlarger or printer exposure after the negative exists.

Changing film exposure changes what the negative in the camera would receive. That can alter density distribution, grain response, halation behavior, and how the print stage is later driven. Changing print exposure changes how the already-developed negative would expose the paper or print stock. Both can make the final image brighter or darker, but they do it at different physical stages.

2.2 Latent Image and Development

In analog photography, exposed silver-halide crystals in the film negative form a latent image that is not visible yet. Developer chemistry turns that latent image into “density”. More exposure generally produces more density, but the relationship is not a perfect straight line across the whole range.

A film or print material has three broad response regions:

- **Toe**: the low-exposure region where shadows begin to separate slowly.
- **Straight line**: the midrange where exposure changes translate most predictably into density changes.
- **Shoulder**: the high-exposure region where additional exposure compresses.

Digital colorists often talk about lift, gamma, gain, contrast, pivot, and highlight rolloff. A film lab thinks in exposure, development, print timing, paper response, and scan interpretation. spektrafilm translates many of those lab decisions into controls you can operate in Resolve.

2.3 Density and Negative Logic

Density is the amount of light a developed material blocks. Higher density transmits less light. In a color negative, the dye image is complementary: the negative is (generally) not meant to be viewed as the final positive image. It is an intermediate that controls how much red, green, and blue-sensitive exposure reaches the print material.

This is why some controls feel inverted compared with RGB grading. More printer exposure through a negative can make the final print darker. A cyan enlarger filter reduces red exposure at the print stage. A paper density change may affect rendered shadows because dense paper regions become dark after scanning.

2.4 CMY Dye Layers and Color Separation

Color film is built from layered records. After development, those records become cyan, magenta, and yellow dye densities. The final color is not created by adding red, green, and blue values into a pixel, but rather it emerges from how spectral light is absorbed by those dyes, then printed, and scanned.

Color separation is therefore a physical behavior, not only a saturation multiplication. Stock choice, spectral upsampling, camera filtration, DIR couplers, bleach bypass, print paper, printer lights, and scanner interpretation all change how colors separate.

2.5 Development Contrast: Push and Pull

Push processing means developing more aggressively than normal. It usually increases contrast and apparent speed, and it can make grain and color separation more pronounced. Pull processing is gentler and tends to reduce contrast.

spektrafilm exposes this separately for film and print:

- `Film Push / Pull Stops` changes negative development behavior before print exposure, grain, and scan-negative output inherit it.
- `Print Push / Pull Stops` changes paper or print development after print exposure, without changing the negative grain structure.

The `Gamma` controls are advanced direct multipliers on the density-curve slope. Use push/pull first when you want a lab-like process move. Use `Gamma` when you intentionally want a lower-level density-curve adjustment.

2.6 Printing and Printer Timing

The print stage places the developed negative back into a light path. The printer or enlarger exposes a print material through that negative. spektrafilm models this as spectral film density, print exposure, optional filtration or APD printer density timing, paper raw exposure, and paper development.

There are two print-timing approaches:

- `Filtered Enlarger` uses spectral CMY filtration controls: `C Filter`, `M Filter Shift`, and `Y Filter Shift`.
- `Printer Density` exposes printer-point controls based on Academy Printing Density responsivities: `Printer Point R`, `Printer Point G`, and `Printer Point B`.

One printer point is equal to 1/12 stop EV. Positive printer points mean more exposure in that printer channel.

2.7 Preflash and Highlight Texture

Preflash gives the print material a small uniform exposure before the image exposure. In practice, it can affect the toe of the print response, reveal highlight information, and soften the harshness of pure whites.

In spektrafilm, `Preflash Exposure` adds this paper-stage pre-exposure. `Preflash M Filter Shift` and `Preflash Y Filter Shift` color only the preflash component, so the affected toe can carry a subtle cast without tinting the whole image in display space.

2.8 Grain, Halation, and Diffusion

Film grain is part of the developed image structure. It is not just noise laid over the final RGB frame. spektrafilm's grain models operate in or around film density, then flow through print and scan stages.

Halation is different from the effect you get from camera diffusion filters. It comes from light scattering through the emulsion and reflecting back from the film base, often with a red bias around strong highlights. Diffusion behaves more like optical softening from filters or print-stage optics. Both can glow, but they originate at different points in the model.

`Film Format` drives the scaling of these spacial effects. It defines the physical film gate used by grain, halation, diffusion, DIR spread, and scanner optics. Changing output resolution changes sampling density; changing film format changes the simulated physical scale.

2.9 Scanning and Display Output

After the film or print density is developed, it has to become an RGB signal. The scanner stage interprets spectral density through viewing light, color matching functions, scanner correction, glare, blur, and unsharp masking before then reaching the final output encoding.

The final `Output Role` determines where the result goes:

- `Display Out SDR` writes a finished SDR display-referred image.
- `Display Out HDR` writes a finished Rec.2100 HDR signal using the chosen transfer, reference white, peak, exposure trim, and tone mapping.
- `Scene Handoff` writes a scene-referred working-space output for continued grading or downstream CST. It is currently still in development

3 Analog Lab To Digital Grade

This section maps common color-grading goals to film-lab concepts and spektrafilm controls.

3.1 Start With Stage Order

The same visible result can be produced in several places, but each place has a different meaning.

Goal	**Film-lab idea**	**spektrafilm controls**
Set technical color	Know what image enters and leaves the lab	Input Color Space, Output Role, Output Color Space, HDR controls
Place exposure	Expose the negative or expose the print	Film Exposure EV, Print Exposure EV, Auto Exposure
Balance color	Time the print or filter the enlarger	C Filter, M Filter Shift, Y Filter Shift, Printer Point R/G/B
Change contrast	Change development or paper response	Film Push / Pull Stops, Print Push / Pull Stops, Gamma, Shadow Shape, Highlight Shape
Change color separation	Change stock chemistry and interlayer behavior	Stock, Paper, DIR Couplers, Negative Bleach Bypass, Print Bleach Bypass
Change texture	Change physical format, grain, optics, and scan	Film Format, Grain, Halation, Diffusion, Scanner

3.2 Balancing

For basic shot matching, start upstream in Resolve. Correct obvious camera white-balance errors, exposure problems, or scene-to-scene mismatches before spektrafilm when those errors are not meant to become part of the film process.

Creative balancing can be achieved in spektrafilm either via CMY filter adjustments or RGB printer lights:

- Use `Filtered Enlarger` when you want CMY enlarger logic. `C Filter` reduces red print exposure, `M Filter Shift` trims magenta filtration, and `Y Filter Shift` trims yellow filtration.

- Use `Printer Density` when you want printer-point timing. `Printer Point R/G/B` are measured in 1/12-stop points. `Gang Printer Points` makes all three channels the same value; `Group Printer Points` preserves offsets while moving all channels by the same delta.

The printer controls are not RGB lift/gamma/gain. They change print exposure channels before paper development and scan.

3.3 Exposure

Use Film `Exposure EV` when you want to change how the negative is exposed. This is closest to changing camera exposure or rating the stock differently. It affects what part of the film response curve the scene lands on and can change how grain, halation, and later print response feel.

Use Print `Exposure EV` when the negative feels right but the print needs to be timed lighter or darker. This is closer to changing the enlarger or printer exposure after the negative already exists.

Use `Auto Exposure` cautiously. It meters decoded source luminance and applies film exposure so the metered value lands near the reference target. This can be useful for fast setup, but for a locked look or manual show workflow, fixed exposure is easier to reason about.

3.4 Contrast

Choose the contrast control by asking which stage should change:

- For negative contrast and a pushed-camera feeling, use `Film Push / Pull Stops`.
- For print-paper contrast, use `Print Push / Pull Stops`.
- For advanced direct curve slope changes, use Film or Print `Gamma`.
- For rendered toe and shoulder shaping, use `Shadow Shape` and `Highlight Shape`.

`Shadow Shape` and `Highlight Shape` operate through the print density response. Positive `Shadow Shape` opens rendered shadows while preserving the paper endpoints. Negative `Highlight Shape` gives gentler highlights and a more pulled-back shoulder.

3.5 Saturation and Color Separation

spektrafilm does not expose a main saturation knob because the model has more physically meaningful ways to shape color.

For broad color personality, change `Stock` and `Paper`. For more chemical color separation, use `DIR Couplers`; increasing `Amount` strengthens the interlayer redevelopment behavior. For silver-retention looks, use `Negative Bleach Bypass` or `Print Bleach Bypass`, understanding that the current implementation is a modeled extension rather than measured stock-specific bleach data.

For camera-side spectral behavior, use `Filtering`. UV and IR cuts affect the light that reaches the film model before exposure; they are not display-space tint controls.

3.6 Highlight and Shadow Behavior

For clipped-looking or hard highlights, check whether you are overdriving the film, print, scanner, or final HDR mapping. Those are different problems.

- Film highlight compression comes from exposure placement and stock response.
- Print highlight shape comes from paper response, print exposure, `Preflash Exposure`, and `Highlight Shape`.
- Softer black floor usually belongs to `Shadow Shape`, scanner glare, or upstream grading, depending on the cause.
- HDR highlight handling belongs to `HDR Tone Mapping`, `Peak Nits`, `Reference White Nits`, and `HDR Exposure EV`.

3.7 Texture and Optical Character

Use `Film Format` early when designing texture. Smaller formats make the same grain and optical spread resolve larger in the final image. Larger formats feel finer because the image samples a larger physical negative. The selected format produces effects resolution-independantly, 1080p and 2160p projects sample the same “digital negative”, just at different densities, resulting in very close resemblance between all output resolutions.

Use `Grain` for density texture, `Halation` for red-biased film-base highlight return, `Diffusion` for camera or print optical softening, and `Scanner` for scan-stage glare, resolving power, blur, and unsharp masking.

Camera diffusion and print diffusion are not interchangeable. Camera diffusion happens before film exposure and can influence what the negative receives. Print diffusion happens in the print path after the negative exists.

3.8 Where To Grade Around spektrafilm

Use upstream Resolve nodes for:

- exposure matching between shots;
- scene-level white balance;
- power windows and corrections that should happen before the film process;
- creative color shaping that should feed the negative.

Use spektrafilm for creative look creation leveraging:

- stock and paper options;
- film and print exposure relationship;
- lab timing, push/pull, preflash, bleach bypass, DIR behavior;
- grain, halation, diffusion, scanner, and final output role.

Use downstream nodes only when your output role is designed for it. If spektrafilm is set to `Display Out SDR` or `Display Out HDR`, treat it as the final display rendering node unless you have a deliberate finishing reason. If it is set to `Scene Handoff`, downstream grading and CST work are expected.

4 Controls Reference

spektrafilm's controls are grouped by process stages. This reference describes the important controls by what part of the spectral film model they affect, what changing them usually does, and in some cases what to watch out for.

spektrafilm OFX comes in different editions. The base spektrafilm version includes a host of options for finetuning specific looks, while spektrafilm\flow is intended for colorists wanting to build looks quickly. Edition flags show which plugin variants the controls show up in: Flow/Film means both the standard spektrafilm as well as `flow` include it; Film only means it is only exposed in the main `spektrafilm` version.

4.1 Color Management

This group defines the technical contract between Resolve and spektrafilm.

- `Mode` Flow/Film : `Print Simulation` renders a positive print path. `Scan Negative` stops at the developed film density and scans the negative directly.
- `Input Color Space` Flow/Film : Must match the data arriving at the OFX node.
- `Output Role` Flow/Film : Chooses the final output policy. `Display Out SDR` writes a finished SDR display image. `Scene Handoff` writes a scene-referred working-space result. `Display Out HDR` writes a finished Rec.2100 HDR result.
- `Output Color Space` Flow/Film : Appears for SDR and Scene Handoff roles. In SDR, choose a display-referred target such as Rec.709 Gamma 2.4. In Scene Handoff, choose the scene-referred working space for downstream color management.
- `HDR Preset` Flow/Film , `HDR Transfer` Flow/Film , `Reference White Nits` Flow/Film , `Peak Nits` Flow/Film , `HDR Exposure EV` Flow/Film , `HDR Tone Mapping` Flow/Film : Define the HDR container and final display mapping. `Hard Clip` preserves calibrated headroom to the chosen peak; `Soft Rolloff` creates a gentler creative shoulder.

Color-management caveat

Do not stack an additional display transform after `Display Out SDR` or `Display Out HDR` unless you are intentionally doing so. For downstream CST workflows, use `Scene Handoff`.

4.2 Filtering

Filtering is camera-side spectral filtration before film exposure.

- `Filter UV` Film only and `UV Cut nm` Film only : Remove ultraviolet/violet energy before it reaches the film model. Lower cut values preserve more violet.
- `Filter IR` Film only and `IR Cut nm` Film only : Remove deep red/infrared energy before exposure. Higher cut values preserve more deep-red contribution.

These controls are useful when exploring camera filtration or unusual spectral looks. They are not white-balance or tint controls.

4.3 Film Plane

The Film Plane group defines how the source image maps to the physical negative.

- `Scale` Film only : Magnifies the sampled film area. Higher values show a smaller crop of the film plane, making grain and optical effects resolve larger.
- `Offset X %` Film only and `Offset Y %` Film only : Pan the enlarged crop across the film plane.

Use these controls for physical framing and enlargement behavior (e.g. cropping into the film negative), not for general output scaling.

4.4 Film

The Film group controls camera exposure, stock choice, negative development, and film-density behavior.

- `RGB to Raw` Flow/Film : Selects the spectral upsampling method before film exposure. `Hanatos 2025` is the default modern path; `Mallett 2019` is available as an alternate method.
- `Stock` Flow/Film : Selects the film profile. This affects spectral sensitivity, density curves, grain, and stock-dependent defaults.

- `Film Format` Flow/Film : Sets the shared physical film gate for grain, halation, diffusion, DIR spread, and scanner optics.
- `Exposure EV` Flow/Film : Camera-side exposure before density formation. Use this to place the scene on the film response.
- `Auto Exposure` Film only and `Auto Exposure Meter` Film only : Meter decoded source luminance and apply a film exposure correction. Useful for setup, less ideal for locked manual timing.
- `Push / Pull Mode` Film only : `Standard` uses the stable timing/gamma model. `Experimental` uses tone-region and layer-dependent development warping.
- `Film Push / Pull Stops` Flow/Film : Positive values push the negative for more contrast and effective speed; negative values pull for softer development.
- `Negative Bleach Bypass` in development : Adds modeled retained silver behavior before print exposure or negative scan.
- `Leuco-Cyan Coupling` in development : Advanced multiplier for the negative bleach-bypass cyan-loss component. Keep at `1.0` unless intentionally researching the effect.
- `Gamma` Film only : Advanced density-curve slope multiplier. Prefer push/pull for normal lab-style moves.

4.5 Print

The Print group controls paper or print stock, print exposure, print development, timing, preflash, and printer-point behavior.

- `Paper` Flow/Film : Selects the print material profile.
- `Print Timing` Flow/Film : `Filtered Enlarger` exposes CMY filtration controls. `Printer Density` exposes APD printer-point controls.
- `Exposure EV` Flow/Film : Enlarger or printer exposure applied to the print material.
- `Print Push / Pull Stops` Flow/Film : Changes print or paper development contrast after print exposure.
- `Print Bleach Bypass` in development : Adds modeled retained silver after print development and before scan.
- `Gamma` Film only : Advanced print density-curve slope multiplier.

- `Shadow Shape` Flow/Film : Shapes rendered shadows through paper density. Positive values open shadows while preserving endpoints.
- `Highlight Shape` Flow/Film : Shapes rendered highlights through paper density. Negative values produce gentler highlight rolloff.
- `C Filter` Flow/Film , `M Filter Shift` Flow/Film , `Y Filter Shift` Flow/Film : CMY enlarger filtration controls for `Filtered Enlarger` mode.
- `Preflash Exposure` Film only : Adds uniform paper pre-exposure to affect the toe and soften highlights.
- `Preflash M Filter Shift` Film only , `Preflash Y Filter Shift` Film only : Color only the preflash component.
- `Gang Printer Points` Flow/Film : Links printer points so one edit sets R, G, and B to the same value.
- `Group Printer Points` Flow/Film : Links printer points so one edit adds the same delta while preserving offsets.
- `Printer Point R/G/B` Flow/Film : APD printer-light trims in 1/12-stop printer points.
- `Printer Point Calibration` Film only : Applies generated neutral calibration offsets for APD printer timing.

4.6 DIR Couplers

DIR couplers model developer-inhibitor interactions that change same-layer contrast and interlayer color behavior.

- `Amount` Flow/Film : Global DIR strength. `0.0` bypasses the effect; `1.0` is the Python-normal strength.
- `Diffusion um` Flow/Film : Lateral spread of inhibitor behavior in the film plane.
- `Same-Layer Inhibition` Flow/Film : Shapes same-layer density behavior, primarily contrast.
- `Interlayer Inhibition` Flow/Film : Shapes cross-layer behavior, primarily color separation.
- `Same-Layer Gamma RGB` Film only , `R -> G/B Gamma` Film only , `G -> R/B Gamma` Film only , `B -> R/G Gamma` Film only : Advanced per-layer terms.

- `Calibrate to Stock` Film only : Restores stock-derived DIR coefficients and resumes stock-follow calibration.

Use the simple controls first. The gamma matrices are research-level controls and can quickly create unstable or non-photochemical behavior if pushed far.

4.7 Grain

The Grain group controls density-domain film grain.

- `Enabled` Flow/Film : Master grain switch.
- `Model` Flow/Film : `Preview` is optimized for interactive work. `Production` uses layered dye-cloud grain. `Grain Synthesis` (in development) rebuilds density from a stochastic Boolean grain field and is much heavier.
- `Sublayers` Flow/Film and `Sub Layer Count` Film only : Split production grain into emulsion sublayers.
- `Particle Area um2` Flow/Film , `Particle Scale RGB` Film only , `Layer Scale` Film only : Control physical particle size and per-layer/channel scaling.
- `Density Min` Film only , `Uniformity RGB` Film only : Control minimum density contribution and clumping/evenness.
- `Final Grain Blur` Film only , `Dye Cloud Blur um` Film only , `Micro Structure` Film only : Shape the optical softness and fine structure of grain. Final Grain Blur is 35mm-equivalent and is automatically compensated for the selected film format.
- `Seed` Film only and `Animate` Film only : Control repeatability and frame-to-frame grain motion.
- `Synthesis Size` in development , `Synthesis Amount` in development , `Synthesis Sharpness` in development , `Synthesis Quality` in development : Simplified controls for Grain Synthesis when that model is active.

Use `Preview` while building the look, then switch to `Production` or `Grain Synthesis` only when the added fidelity is worth the render cost.

4.8 Grain Synthesis

This collapsed group exposes research controls for the Boolean grain synthesis model.

- `Samples` in development , `Mean Radius um` in development , `Radius StdDev Ratio` in development , `Observation Aperture Sigma um` in development , `Cell Size Ratio` in development , `Max Radius Quantile` in development , `Coverage Epsilon` in development , `Max Grains Per Cell` in development : Control the Monte Carlo grain estimator and its physical sampling.
- `Radius Scale RGB` in development , `Layer Scale` in development , `Layered` in development : Control channel and depth-layer scaling.

These controls are intended for final-quality tests and research. For normal grading, use the simplified Grain group controls.

4.9 Halation

Halation models film-base scatter and emulsion back-reflection before development.

- `Enabled` Flow/Film : Master halation/scatter switch.
- `Scatter Amount` Film only , `Scatter Scale` Film only : Near-emulsion scatter strength and radius.
- `Amount` Flow/Film , `Scale` Flow/Film : Red-biased film-base halation strength and radius.
- `Strength RGB` Flow/Film : Per-channel weighting for returned halation light.

Halation is most visible around strong highlights and high-contrast edges. If the whole image feels soft, reduce scatter or scale before assuming the issue is print diffusion.

4.10 Diffusion

Diffusion models camera-stage or print-stage optical diffusion.

- `Camera Enabled` Flow/Film , `Camera Family` Flow/Film , `Camera Strength` Flow/Film : Apply a diffusion profile before film exposure, similar to a lens filter.
- `Print Enabled` Flow/Film , `Print Family` Flow/Film , `Print Strength` Flow/Film : Apply diffusion in the print path.
- `Camera/Print Spatial Scale` Film only : Radius multiplier for the chosen diffusion family.
- `Camera/Print Halo Warmth` Flow/Film : Warm or cool bias for the diffusion halo.

- `Core Intensity/Size` Film only , `Halo Intensity/Size` Film only , `Bloom Intensity/Size` Film only : Advanced components of the diffusion profile.

Camera diffusion interacts with film exposure. Print diffusion happens after the negative exists. Use that distinction to choose the stage.

4.11 Scanner

The Scanner group controls scan-stage correction and optics after the film or print density exists.

- `Enabled` Flow/Film : Master scanner finishing switch.
- `White Correction` Flow/Film , `White Level` Flow/Film : Normalize scan white when enabled.
- `Black Correction` Flow/Film , `Black Level` Flow/Film : Normalize scan black when enabled.
- `Glare Percent` Film only : Adds scanner veiling glare, often lifting shadows subtly.
- `MTF50 lp/mm` Film only : Scanner resolving power. Higher values preserve finer detail; `0` bypasses optical blur.
- `Unsharp Radius um` Film only , `Unsharp Amount` Film only : Film-plane unsharp masking.

Use scanner controls as finishing behavior, not as the first place to fix a bad input transform.

4.12 Manage

The Manage group handles LUT export, parameter transfer, and startup defaults.

- `LUT Size` Flow/Film : 33 or 65 point cube resolution.
- `LUT Destination` Flow/Film : Application-oriented export folder preset.
- `LUT Identifier` Flow/Film : Filename name segment.
- `Export LUT` Flow/Film : Exports a Display Out SDR color-only `.cube` LUT.
- `Copy Params` Flow/Film : Copies current visible Spektra settings to the system clipboard.
- `Paste Params` Flow/Film : Applies copied Spektra settings to matching controls.

- `Set Defaults` Flow/Film : Saves the current settings as startup defaults for new Spektra instances.
- `Reset Factory Defaults` Flow/Film : Deletes saved user defaults and resets current controls to factory values.
- `Open User Manual` Flow/Film : Opens the bundled PDF manual from the installed OFX bundle.

> **LUT export limit**
> LUT export is available only for `Display Out SDR`, and it contains the color-only spectral transform. Grain, halation, diffusion, film-plane transforms, scanner blur/sharpening, and other spatial or stochastic effects are not represented in the LUT.

4.13 Info

The Info group displays plugin version, creator credit, and foundational research credit.

5 Practical Workflows

This section gives repeatable starting workflows. Treat the values as direction, not as universal presets. The correct amount depends on footage, timeline color management, output target, and the stock/paper combination.

5.1 Basic Resolve Setup

1. Put any camera RAW settings, CSTs, exposure matching, and primary cleanup before spektrafilm.
2. Add spektrafilm near the end of the node tree.
3. Set `Input Color Space` to the actual data entering the node.
4. Choose `Output Role`.
5. If using `Display Out SDR`, set `Output Color Space` to the final SDR display target, commonly Rec.709 Gamma 2.4.
6. If using `Scene Handoff`, choose the downstream working space and finish with a correctly configured CST or color-management output.
7. If using `Display Out HDR`, set the HDR preset, transfer, reference white, peak nits, exposure trim, and tone mapping for the mastering target.

5.2 Node Placement

For most grades, spektrafilm should behave like a print or finishing node. Keep shot-level balance upstream. Then use spektrafilm for stock, print, process, texture, and display rendering.

Use downstream nodes sparingly when `Display Out SDR` or `Display Out HDR` is active. Those modes are finished display outputs. Downstream display transforms can double-tone-map or distort the result.

5.3 Scene Handoff With Resolve CST

Use `Scene Handoff` when the image needs to remain in a scene-referred working space after spektrafilm. This is useful when a facility pipeline expects a later CST, a show-level transform, or additional scene-referred finishing.

Recommended setup:

1. In spektrafilm, set `Output Role` to `Scene Handoff`.

2. Set `Output Color Space` to the scene-referred space expected downstream, such as DaVinci Intermediate WideGamut.
3. Add a downstream Resolve CST or color-management output node.
4. In the downstream CST, turn tone mapping off.
5. Enable inverse OOTF when needed by the downstream transform so spektrafilm's display-intent scan rendering is handed back into the scene-referred chain correctly.

If the result becomes very high contrast, muddy, or clipped, check for double tone mapping first.

5.4 HDR Output

Use `Display Out HDR` when spektrafilm should produce the finished HDR display signal.

- `HDR Preset`: Start with the preset closest to the delivery target.
- `HDR Transfer`: Use PQ for HDR10/Dolby Vision style mastering, HLG for broadcast-style HLG deliverables.
- `Reference White Nits`: Defines the luminance assigned to relative scan white.
- `Peak Nits`: Match the mastering target or monitor.
- `HDR Exposure EV`: Final HDR brightness trim after scan.
- `HDR Tone Mapping`: Use `Hard Clip` for calibrated peak behavior, `Soft Rolloff` for a gentler creative shoulder.

Do not feed Display Out HDR into another scene-to-display transform unless the downstream pipeline explicitly expects that.

5.5 Recipe: Neutral Setup For A New Shot

1. Match the shot upstream in Resolve before spektrafilm.
2. Set `Input Color Space` correctly.
3. Choose `Print Simulation`.
4. Leave `Film Push / Pull Stops`, `Print Push / Pull Stops`, bleach bypass, preflash, DIR, diffusion, and scanner correction near defaults.
5. Choose the intended `Stock`, `Paper`, and `Film Format`.
6. Adjust Film `Exposure EV` for negative placement.
7. Adjust Print `Exposure EV` for final print timing.

8. Only after the image is technically stable, add grain, halation, diffusion, DIR, bleach bypass, or scanner finishing.

5.6 Recipe: Warmer Or Cooler Print Balance

For CMY enlarger timing:

1. Set `Print Timing` to `Filtered Enlarger`.
2. Use `M Filter Shift` and `Y Filter Shift` for practical print-balance moves.
3. Recheck Print `Exposure EV`, because color filtration can change perceived density.

For printer-point timing:

1. Set `Print Timing` to `Printer Density`.
2. Use `Printer Point R/G/B` in small moves.
3. Use `Group Printer Points` when you want to preserve color offsets while changing overall density.
4. Use `Gang Printer Points` when you want all points set to one neutral value.

5.7 Recipe: Softer Highlights With Preflash

1. In the Print group, raise `Preflash Exposure` from zero in small increments.
2. If the softened highlights need color, trim `Preflash M Filter Shift` or `Preflash Y Filter Shift`.
3. Use `Highlight Shape` for additional rendered-highlight shaping.
4. Avoid compensating too aggressively with RGB gain downstream; the point of preflash is that the toe is affected through the paper response.

5.8 Recipe: Stronger Print Contrast

1. Increase `Print Push / Pull Stops` for paper-stage contrast.
2. Fine tune with Print `Gamma` only if the push/pull control is not specific enough.
3. Use `Highlight Shape` to decide whether highlights should stay firm or roll off more gently.
4. Use `Shadow Shape` to keep deeper blacks readable if the print becomes too heavy.

5.9 Recipe: Pushed Negative Look

1. Increase `Film Push / Pull Stops`.
2. Rebalance Film `Exposure EV`; pushed development can change where midtones feel correctly placed.
3. Check grain at final resolution, because pushed negative behavior can make texture feel stronger.
4. If color separation becomes too intense, reduce DIR Couplers `Amount` or pull back print contrast.

5.10 Recipe: Retained-Silver Bleach-Bypass Look (in development)

1. Start with small `Negative Bleach Bypass` values if you want the retained-silver behavior to affect print exposure and scan-negative output.
2. Use `Print Bleach Bypass` if the silver-retention feel should happen after print development.
3. Rebalance Print `Exposure EV` and print contrast after adding bypass.
4. Treat `Leuco-Cyan Coupling` as an advanced research control; leave it at `1.0` unless intentionally testing the model.

The current bleach-bypass behavior is a physically motivated model extension. It requires more stock-specific measurements to be production ready.

5.11 Recipe: Cleaner Modern Print

1. Use a larger `Film Format`, such as 35mm, 65mm, or IMAX-style formats, for finer physical texture.
2. Keep `Halation Amount` and `Diffusion Strength` low or disabled.
3. Use modest print contrast and minimal preflash.
4. Keep scanner glare low and scanner unsharp moderate.

5.12 Recipe: Softer Diffusion And Halation

1. Use `Halation Amount` for highlight-edge return light and red-biased glow.
2. Use `Scatter Amount` carefully; too much can soften the whole image.
3. Use `Camera Enabled` diffusion when the glow should affect the negative exposure.

4. Use `Print Enabled` diffusion when the softness should feel like a print-path optical effect.
5. Adjust `Halo Warmth`, core, halo, and bloom components only after the basic strength and scale feel correct.

5.13 LUT Export Workflow

1. Set `Output Role` to `Display Out SDR`.
2. Choose the SDR `Output Color Space` for the LUT target.
3. Choose `LUT Size`. Use 65 for higher quality, 33 for smaller files and faster export.
4. Choose `LUT Destination` and set `LUT Identifier` (Itentifiers are displayed in the filename. The LUT naming pattern is: YYMMDD_identifier_randomString).
5. Press `Export LUT`.

The exported LUT is color-only. It will not include grain, halation, diffusion, film-plane transforms, scanner blur, scanner sharpening, or animated/stochastic effects. For final quality, use the live OFX plugin or render through Resolve.

5.14 Defaults And Parameter Transfer

Use `Copy Params` and `Paste Params` to move settings between spektrafilm instances (e.g. from spektrafilm to spektrafilm flow). Use `Set Defaults` only when the current settings should become the default for newly created instances. `Reset Factory Defaults` deletes saved user defaults and resets the current instance to factory values. Use `Open User Manual` to open the PDF that ships inside the OFX bundle.

Saved defaults are useful for a show baseline, but they are not a replacement for project-level versioning. Record important show looks in Resolve stills, PowerGrades, or project documentation as well.

6 Technical Handbook: Practical Model

This handbook section explains spektrafilm as a working model for advanced colorists. It stays close to grading decisions: what each stage receives, what it produces, which controls belong there, and what problems should not be solved there.

6.1 Reading The Pipeline As A Colorist

Most digital grades operate directly on RGB image values. spektrafilm is different: many controls act before the final RGB image exists. The process is closer to a chain of materials and optical operations:

1. Decode the incoming RGB image into scene-linear light.
2. Convert RGB light into a film raw exposure estimate.
3. Apply camera-side filtration, film-plane transforms, halation, scatter, and camera diffusion.
4. Develop film raw exposure into CMY film density.
5. Apply DIR coupler behavior and grain in density space.
6. Print the negative onto a paper or print stock.
7. Develop print raw exposure into print density.
8. Scan the film or print density back into RGB.
9. Apply scanner finishing and output encoding.

6.2 Input Decoding And Working Space

`Input Color Space` tells spektrafilm how the video stream arriving at the OFX node. It lets spektrafilm correctly convert the input to linear space, similar to a CST. If this is wrong, every later stage is wrong: spectral upsampling sees the wrong colors, film exposure lands in the wrong range, and print density reacts to a false negative.

In short:

- In a Resolve Color Management timeline, set `Input Color Space` to the timeline data entering the node.
- In a node-based CST workflow, set it to the output of the previous node.
- If the footage is already display-referred Rec.709, choose the matching Rec.709 output.

The plugin is most predictable when it receives wide-gamut scene-referred or log-derived data with highlights preserved. Clipped display material can still be styled, but it cannot recover film-like highlight information that is no longer present.

6.3 Spectral Upsampling

Analog film does not respond to three RGB primaries. It responds to spectral energy across wavelengths. `RGB to Raw` chooses the method used to infer a plausible spectrum or raw film exposure response from RGB source data.

`Hanatos 2025` is the default path and should be the normal choice. `Mallett 2019` is available as an alternate spectral method. These are not look presets; they are different interpretations of how RGB data should become film-relevant spectral exposure.

Practical consequences:

- Saturated LED, neon, and synthetic colors may react differently between methods.
- Camera filtration and stock sensitivity matter because the model is not only display RGB.
- If a shot has strange saturated color behavior, compare `RGB to Raw` before trying to fix it with final RGB saturation.

6.4 Camera Filtration And Film Plane

The Filtering group acts before film exposure. `Filter UV`, `UV Cut nm`, `Filter IR`, and `IR Cut nm` change the spectral range that reaches the film model. This is useful for camera-side filtration and unusual spectral behavior, but it is not a temperature/tint replacement.

The Film Plane group sets physical sampling. `Scale` and `Offset X/Y %` determine how the source image maps onto the film gate. The important idea is that spatial effects are measured in film-plane units. `35mm` develops the film emulation on a 35mm film strip and samples that film strip at the project's resolution. At 1080p every horizontal pixel represents ca. 18um on the filmstrip. When you magnify the film plane, grain and optical spread resolve differently because the same physical structure is sampled at a different scale.

Use Film Plane controls when the physical geometry is part of the look. Do not use them as a general layout or resize tool.

6.5 Film Exposure

Film `Exposure EV` applies before density formation. This is analogous to camera exposure or rating the stock. It decides where the scene lands on the negative response.

Use Film `Exposure EV` when:

- highlights should enter the shoulder differently;
- shadows should sit deeper or higher on the negative;
- grain, halation, and print response should inherit a different negative;
- a scene should feel underexposed or overexposed before printing.

Avoid using Film `Exposure EV` for:

- final deliverable brightness;
- print timing after the negative already feels right;
- compensating a wrong `Input Color Space`.

`Auto Exposure` is a setup tool. It meters decoded source luminance and applies film exposure to land near the reference target. It is useful for quickly normalizing material or counteracting changing lighting conditions, but manual fixed exposure is better for look continuity.

6.6 Film Development

`Film Push / Pull Stops` changes negative development. Push generally increases effective speed and contrast; pull softens contrast. In spektrafilm, this happens before print exposure and before scan-negative output.

Use film push/pull when:

- the negative should feel more aggressive;
- grain and color separation should inherit the process;
- the look should resemble a process change rather than a final contrast curve.

Use Film `Gamma` only when you intentionally want a direct density-curve slope multiplier. It is an advanced control and less communicative than push/pull in normal grading notes.

6.7 Negative Bleach Bypass

`Negative Bleach Bypass` applies retained-silver behavior before print exposure or negative scan. It changes the negative that the print stage sees. That is why it can alter contrast, saturation behavior, and density more fundamentally than a post-scan desaturation.

Use it sparingly. Small amounts can add density and severity. Large amounts can quickly make the image heavy, cold, or hard to time. `Leuco-Cyan Coupling` is an advanced research multiplier; the default `1.0` is the intended path.

Because this model is a physically motivated extension rather than measured process data for each stock, document its use as a creative model when building show notes.

6.8 DIR Couplers

DIR couplers are best understood as color-separation and density-interaction controls. They are not merely a saturation slider. In practical use:

- `Amount` controls global strength.
- `Diffusion um` controls the lateral scale of inhibition.
- `Same-Layer Inhibition` primarily changes layer contrast.
- `Interlayer Inhibition` primarily changes cross-layer color separation (saturation).
- Advanced gamma matrices should usually remain stock-calibrated. They tune the behavior between color channels

Start with `Amount`. If the image gains useful color separation, keep it. If it becomes edge-crunchy or chemically implausible, reduce it. Use `Calibrate to Stock` when advanced terms have drifted.

6.9 Grain

spektrafilm grain is connected to film density and film-plane geometry. This is why `Film Format`, `Scale`, and output resolution affect how grain feels.

Use the grain models this way:

- `Preview`: look-building and interactive grading.
- `Production`: final-quality density-domain dye-cloud grain.

- `Grain Synthesis`: heavy research/final-render model that reconstructs density from a stochastic grain field.

For normal grading, choose format first, then grain model, then amount/size/blur controls. Do not judge grain at a viewer zoom level that will not be used for delivery.

6.10 Print Exposure

Print `Exposure EV` happens after the negative exists. It is closer to enlarger/printer exposure than camera exposure.

Use Print `Exposure EV` when:

- the negative feels correct but the print needs to be lighter or darker;
- print contrast and paper response should stay the main look driver;
- grouped printer points need a neutral density adjustment.

If changing Print `Exposure EV` destroys highlight or shadow intent, the negative may be placed poorly. Go back to Film `Exposure EV`, film push/pull, or stock choice.

6.11 Print Timing

`Print Timing` selects how print exposure color is controlled.

`Filtered Enlarger` exposes CMY filtration:

- `C Filter` reduces red exposure on the print.
- `M Filter Shift` changes magenta filtration relative to neutral.
- `Y Filter Shift` changes yellow filtration relative to neutral.

`Printer Density` exposes APD printer points:

- `Printer Point R/G/B` are measured in 1/12-stop points.
- `Gang Printer Points` forces all channels to the same value.
- `Group Printer Points` preserves offsets while applying the same delta.
- `Printer Point Calibration` keeps generated neutral offsets active.

Choose `Filtered Enlarger` when the creative language is CMY darkroom filtration. Choose `Printer Density` when the creative language is lab timing lights.

6.12 Print Development And Paper Shaping

`Print Push / Pull Stops`, Print `Gamma`, `Shadow Shape`, and `Highlight Shape` all live around paper response, but they do not mean the same thing.

- `Print Push / Pull Stops`: process-like paper development contrast.
- Print `Gamma`: advanced density-curve slope multiplier.
- `Shadow Shape`: rendered shadow shaping through paper density.
- `Highlight Shape`: rendered highlight shaping through paper density.

For most work, use Print Push/Pull for broad contrast, then Shadow/Highlight Shape for toe/shoulder feel. Use Gamma for controlled advanced corrections.

6.13 Preflash

Preflash is paper-stage toe control. It gives the print material a small pre-exposure before image exposure, affecting the lower response without acting like a standard highlight rolloff.

Use `Preflash Exposure` when:

- whites are too hard;
- highlight texture needs to appear without flattening midtones;
- the print should feel flashed or softened in the toe (highlights).

Use `Preflash M Filter Shift` and `Preflash Y Filter Shift` when the affected toe needs a color bias. Keep this subtle; strong colored preflash can look like a display-space tint.

6.14 Print Bleach Bypass

`Print Bleach Bypass` applies retained-silver behavior after print development and before scan. It is different from negative bypass because it changes the final print material, not the negative that exposes the print.

Use print bypass when:

- the finished print should gain density and reduced color purity;
- the look should feel like retained silver at the final material stage;
- negative grain and print exposure should remain mostly unchanged upstream.

6.15 Halation, Diffusion, And Scanner

Halation is film-stage light return. It becomes especially visible around hot highlights. If the whole image undesireably softens, reduce `Scatter Amount` and scale before removing halation entirely.

Diffusion can happen at the camera stage or print stage:

- camera diffusion acts like a diffusion filter put in front of the camera's lens and affects what exposes the negative;
- print diffusion acts like a diffusion filter put in front of the scanner capturing the developed print and affects the print path after the negative exists.

Scanner controls are last-stage finishing:

- `Glare Percent` lifts shadows through scan veiling glare;
- `MTF50 lp/mm` controls resolving power;
- `Unsharp Radius um` and `Unsharp Amount` add scanner-style sharpening.

Do not use scanner controls to fix upstream color-management or exposure mistakes. Use them to make the scan feel physical.

6.16 Output Roles

`Display Out SDR` is a final SDR display output. It is the right choice for direct Rec.709 monitoring and simple SDR delivery tests.

`Scene Handoff` returns the post-scan result to a scene-referred working space. Use it when downstream Resolve color management, a CST, or a show transform must remain after spektrafilm. Disable downstream tone mapping and apply an inverse OOTF unless the pipeline explicitly requires something else.

`Display Out HDR` is a final HDR display output. It writes Rec.2100 PQ or HLG according to the HDR controls. Do not feed it into another scene-to-display transform.

6.17 LUT Export

The Manage group can export a Display Out SDR color-only `.cube` LUT. This is useful for compatibility, dailies or camera-monitoring, but it cannot retain spatial or stochastic behavior.

Excluded from LUT export:

- grain;
- halation;
- diffusion;
- film-plane transforms;
- scanner blur and sharpening;
- animated or stochastic effects.

If the look depends on texture or optics, render through the live OFX plugin.

6.18 Practical Diagnosis

Symptom	Likely stage to check
Wrong overall color	Input Color Space, upstream CST, print timing.
Highlights collapse	Input clipping, Film Exposure EV, Highlight Shape, Preflash Exposure, HDR tone mapping.
Shadows block	Film placement, print exposure, Shadow Shape, scanner glare.
Colors feel plastic	Global saturation upstream, DIR Amount, stock/paper mismatch, final RGB trims.
Grain too large	Film Format, Film Plane Scale, grain particle size, viewer zoom.
Glow everywhere	Scatter Amount, diffusion strength, scanner glare.
LUT does not match OFX	Spatial/stochastic effects are excluded from LUT export.

6.19 Recommended Order For Building A Look

1. Confirm color management.
2. Choose stock, paper, and film format.
3. Place film exposure.
4. Time print exposure.
5. Set print contrast and toe/shoulder behavior.
6. Add color timing and DIR separation.
7. Add preflash or bypass only if the look calls for it.
8. Add grain, halation, diffusion, and scanner finishing.
9. Choose output role and delivery transform.

This order keeps decisions physically legible. It also makes the grade easier to explain to another colorist.

7 Technical Handbook: Imaging Science

This section goes deeper into the image-formation ideas behind spektrafilm. It is written for readers who want the model vocabulary: exposure, density, spectral integration, curve shape, printer points, grain statistics, halation kernels, scanner MTF, and display mapping.

It is not a full source-code specification. When a control is a measured model, a generated table, or a provisional creative extension, the distinction matters and is called out.

7.1 Scene-Linear Light And Log Encodings

Camera log values are storage encodings, not light. Before film simulation, the input must be decoded into a scene-linear exposure representation. In practical terms, a log value is first interpreted according to `Input Color Space`, then transformed into a linear working representation.

The core exposure idea is multiplicative:

```
E_out = E_in * 2^EV
```

An EV change is a power-of-two exposure change. `+1 EV` doubles exposure. `-1 EV` halves exposure.

Most film and paper curve operations happen more naturally in log exposure:

```
logE = log10(max(E, epsilon))
```

This is why exposure offsets and development slope changes behave differently. Exposure moves the input along the curve; development changes the curve relationship.

7.2 Spectral Upsampling

An RGB camera or timeline value has three channels. A film emulsion responds to a continuous spectral distribution. spektrafilm therefore needs a bridge from RGB to film-relevant raw exposure.

Conceptually:

```
RGB scene value -> estimated spectrum S(lambda)
film raw channel c = integral S(lambda) * sensitivity_c(lambda) d
lambda
```

Where:

- `lambda` is wavelength;
- `S(lambda)` is the inferred spectral power distribution;
- `sensitivity_c(lambda)` is the stock sensitivity for a film layer/channel;
- the integral is approximated by sampled spectral tables.

`Hanatos 2025` and `Mallett 2019` are different ways to build that bridge. The result is not merely a color transform. It changes how saturated colors, narrowband lights, and stock sensitivities interact.

7.3 Camera Filtration

Camera filtration modifies light before film exposure. A UV or IR cut can be represented as a spectral transmittance term:

```
S_filtered(lambda) = S(lambda) * T_filter(lambda)
```

For UV/IR cuts, `T_filter(lambda)` is near zero outside the passed band and near one inside it, with a transition width. Because this is applied before film raw exposure, it changes what the film layers receive. It is not equivalent to white balance after scanning.

7.4 Density

Density measures how much light a material blocks. Transmission density is commonly defined as:

```
D = -log10(T)
T = 10^-D
```

Where `T` is transmittance. Higher density means less transmitted light. A density of `0.3` transmits about half the light because `10^-0.3` is close to `0.5`.

Color materials have dye densities, often represented as CMY records. The final scan color emerges from spectral absorption, not from simply subtracting RGB values.

7.5 Film Response Curves

Film and paper response curves map log exposure to density. They usually have:

- a toe, where low exposures separate slowly;
- a straight-line region, where slope is more stable;

- a shoulder, where high exposures compress.

Approximate local contrast is the derivative of density with respect to log exposure:

```
gamma_local = dD / d(logE)
```

The user-facing `Gamma` controls act like density-curve slope multipliers. Push/pull controls are more process-oriented wrappers around development behavior and should generally be preferred for creative grading.

7.6 Exposure Shift Versus Development Change

Exposure changes input placement:

```
logE_shifted = logE + log10(2) * EV
```

Development changes the relationship between log exposure and density. In practical terms:

```
D = curve(logE / gamma_effective)
```

The exact implementation can vary by mode, but the conceptual distinction remains:

- exposure moves the scene through the material response;
- development changes the response slope and density buildup.

7.7 Negative And Positive Polarity

A negative is an intermediate. More density in a negative blocks more printer light. Because color negative dyes are complementary, intuitive RGB sign assumptions often fail.

For a print stage:

```
printer_light(lambda) passes through film density
transmitted(lambda) = printer_light(lambda) * 10^-D_film(lambda)
paper exposure = integral transmitted(lambda) *
paper_sensitivity(lambda) d lambda
```

This is why more printer exposure through the negative does not mean “add RGB brightness” in final display space. It changes paper exposure before paper development.

7.8 Printer Points

Printer points are exposure units. In this plugin, one printer point is `1/12` stop:

```
exposure_multiplier = 2^(points / 12)
```

The APD printer-density path applies these as print exposure trims in the print raw stage. Positive values mean more exposure in that printer channel. The final rendered color depends on the negative, paper response, and scan conversion.

`Gang Printer Points` sets all three point values to one number. `Group Printer Points` preserves offsets while applying a common delta. The grouped behavior is useful for density timing after establishing a color offset.

7.9 CMY Filtration

Subtractive print filtration works by removing spectral energy before paper exposure. Cyan filtration reduces red exposure; magenta filtration reduces green exposure; yellow filtration reduces blue exposure.

Conceptually:

```
print_illuminant_filtered(lambda) =
  illuminant(lambda) * T_C(lambda) * T_M(lambda) * T_Y(lambda)
```

This is why CMY filter controls are different from RGB post-balance controls. They alter the exposing light before paper density exists.

7.10 Paper Development And Print Shape

Print paper or print film has its own density curves. Print exposure first produces paper raw exposure, then paper development maps that raw exposure to print density.

`Print Push / Pull Stops` changes paper development behavior. `Shadow Shape` and `Highlight Shape` reshape rendered toe and shoulder behavior through paper density while preserving endpoints. They are not generic lift/gain controls.

Rendered shadows correspond to high paper density. Rendered highlights correspond to low paper density. This inversion is the source of many confusing sign relationships in print-stage controls.

7.11 Preflash

Preflash adds uniform paper raw exposure before image exposure:

```
raw_print_total = raw_print_image + raw_preflash
log_raw_print = log10(max(raw_print_total, epsilon))
```

Because the preflash is added before paper development, it mostly changes the toe. It can reveal highlight detail and soften white onset while leaving much of the midrange less affected than a display-space gain would.

Colored preflash shifts the spectrum of the preflash component, not the whole image.

7.12 DIR Coupler Model

Developer Inhibitor Releasing couplers change how density develops across and within layers. A useful abstraction is a donor/receiver matrix:

```
corrected_density_layer_i =
  density_i - sum_j inhibition[j -> i] * f(density_j, diffusion)
```

Where:

- diagonal terms are same-layer behavior;
- off-diagonal terms are interlayer behavior;
- diffusion controls lateral spread in the film plane;
- gamma terms shape how strongly density in one layer affects another.

The important grading consequence is that DIR changes color separation and layer contrast before the final RGB scan. It is not the same as increasing saturation.

7.13 Bleach Bypass As Modeled Retained Silver

Bleach removes developed silver after dye formation. Bypassing or reducing bleach leaves silver in the image. spektrafilm's current bypass controls are model extensions based on retained-silver behavior, not measured stock/process datasets.

A simplified conceptual model:

```
dye_density = developed CMY dye image
silver_image = layer-derived retained silver estimate
neutral_silver_density = bypass_amount * silver_image
```

```
total_density(lambda) = dye_spectral_density(lambda) +
neutral_silver_density
```

The native model also uses a density-domain black-image overlay, and negative bypass includes a leuco-cyan coupling component. The point is to avoid a fake post-scan desaturation. The effect lives before spectral reconstruction at the negative or print stage.

Use this model honestly:

- `Negative Bleach Bypass` changes the film density before print exposure or negative scan.
- `Print Bleach Bypass` changes print density after print development and before scan.
- The current silver basis is a conservative model, not a measured silver extinction dataset.

7.14 Grain Models

Film grain is spatial density structure. spektrafilm exposes multiple models because there is a tradeoff between speed, physical detail, and stochastic accuracy.

`Preview` is optimized for interaction. It gives a fast grain impression suitable for look-building.

`Production` uses density-domain dye-cloud modeling and layer controls. It is the normal higher-quality path.

`Grain Synthesis` uses an inhomogeneous Boolean grain-field idea: the image is estimated from stochastic grain coverage in the film plane. A simplified form:

```
coverage = probability that grain particles cover the observation
aperture
density ~= -log10(max(1 - coverage, epsilon))
```

The real renderer uses sampled film-plane cells, radius controls, channel/layer scaling, observation aperture, and Monte Carlo samples. The practical controls are `Synthesis Size`, `Synthesis Amount`, `Synthesis Sharpness`, and `Synthesis Quality`; the collapsed Grain Synthesis group exposes deeper research parameters.

7.15 Film Format And Physical Units

Spatial controls use film-plane units. If an effect radius is expressed in micrometers, the same radius will cover a different number of output pixels depending on:

- film format;
- film-plane scale;
- render resolution;
- crop/enlarger behavior.

Conceptually:

```
radius_pixels = radius_um / pixel_size_um
```

This is why `Film Format` matters for grain, halation, diffusion, DIR spread, and scanner optics. It is not only a label.

7.16 Halation And Scatter

Halation is light returning through the emulsion and film base, usually strongest around bright sources and often red-biased. Scatter is a nearer, broader optical redistribution in the emulsion path.

A simplified convolution view:

```
raw_exposure_out =
  raw_exposure_direct
  + amount_scatter * blur(raw_exposure, sigma_scatter)
  + amount_halation * color_weight * blur(highlight_energy,
sigma_halation)
```

The plugin separates amount, scale, and RGB strength controls. If halation behaves like global softness, reduce scatter or scale. If highlight edges lack red return, adjust halation amount and RGB strength.

7.17 Diffusion

Diffusion is modeled as an optical redistribution of light across core, halo, and bloom scales. It can be applied at camera stage or print stage.

Conceptual form:

```
diffused = direct * w0
         + blur(image, sigma_core) * w_core
```

```
          + blur(image, sigma_halo) * w_halo
          + blur(image, sigma_bloom) * w_bloom
```

The family selection controls the profile; strength, spatial scale, halo warmth, and advanced component controls reshape it.

Camera diffusion changes film exposure. Print diffusion changes print exposure. The same numerical strength can feel different because the stage is different.

7.18 Scanner Conversion

Scanning turns film or print density back into RGB. Conceptually:

```
transmitted(lambda) = viewing_illuminant(lambda) * 10^-
D_material(lambda)
XYZ = integral transmitted(lambda) * CIE_1931(lambda) d lambda
RGB = output_matrix * XYZ
```

Scanner correction and optics then shape the result:

- white/black correction remaps reference levels;
- glare adds veiling light;
- MTF controls resolving power;
- unsharp masking adds edge enhancement.

`MTF50 lp/mm` is a resolution metric: the spatial frequency where modulation transfer falls to 50 percent. Higher values preserve finer film-plane detail. A value of zero bypasses optical blur.

7.19 HDR Mapping

HDR output is not just “make it brighter.” spektrafilm scans the material result, then maps it into a finished HDR display signal when `Display Out HDR` is active.

Important controls:

- `HDR Transfer`: PQ or HLG.
- `Reference White Nits`: luminance assigned to relative scan white.
- `Peak Nits`: display peak used by scaling and tone mapping.
- `HDR Exposure EV`: post-scan HDR brightness trim.
- `HDR Tone Mapping`: hard clip or soft rolloff.

Hard clipping preserves calibrated headroom until peak. Soft rolloff compresses highlights creatively near the peak.

Do not put another scene-to-display transform after `Display Out HDR`. If a pipeline needs a downstream output transform, use `Scene Handoff`.

7.20 Scene Handoff

`Scene Handoff` writes the post-scan result into a scene-referred working space for downstream color management. It exists because some Resolve pipelines need the film-rendered image to continue through a show transform or CST.

The common failure mode is double tone mapping. If a downstream CST tone maps a result that spektrafilm already shaped for display intent, highlights can become muddy or overly compressed. Turn downstream tone mapping off unless the show pipeline explicitly requires it.

7.21 LUT Export Limits

A 3D LUT maps input RGB triplets to output RGB triplets. It cannot encode spatial, stochastic, temporal, or resolution-dependent behavior.

Included in the LUT:

- color-only spectral transform for `Display Out SDR`;
- stock/paper color response as sampled through the color-only path;
- compatible SDR output encoding.

Excluded from the LUT:

- grain;
- halation and scatter;
- camera or print diffusion;
- film-plane scale and offset;
- scanner blur, unsharp masking, and spatial glare;
- animation and stochastic sampling.

7.22 Numerical Stability And Clipping

Physical models need floors and guards. Raw exposure cannot be allowed to become negative before `log10`; density conversion needs epsilon floors; output encoding should preserve meaningful overrange when downstream nodes need it.

Practical implications:

- If the input is clipped before spektrafilm, film shoulder behavior cannot reconstruct lost detail.
- If downstream nodes clamp the OFX output, scene handoff and HDR workflows may lose range.
- If a control creates extreme densities, final RGB may become hard to time even if the math is valid.

7.23 Model Honesty

spektrafilm combines measured data, generated tables, physically motivated models, and creative extensions. Use the controls with the correct confidence level.

Higher-confidence areas:

- color-management transforms for supported spaces;
- generated film/paper density curves from profile data;
- spectral scan conversion using illuminants, CIE CMFs, and output matrices;
- printer-point units and stage placement;
- physically staged grain, halation, diffusion, and scanner operations.

Areas to describe as modeled or provisional:

- bleach bypass retained-silver behavior;
- exact stock-specific push/pull process changes where measured alternate curves are unavailable;
- Grain Synthesis as a heavy stochastic research model rather than a bit-identical laboratory reproduction;
- any visual match to a named film look without the original footage, lighting, show LUT, and colorist decisions.

7.24 Control-Stage Summary

Control family	Model stage	Scientific idea
Input/output color	Decode and encode	Transfer functions, primaries, scene/display reference.
Filtering	Before film exposure	Spectral transmittance multiplication.
Film exposure	Raw film exposure	Multiplicative exposure, log placement.

Film push/pull	Film development	Log-exposure-to-density slope and process activity.
DIR	Developed film density	Layer inhibition and color separation.
Grain	Film density structure	Spatial stochastic density variation.
Print exposure/ timing	Paper raw exposure	Spectral transmission through negative and printer light.
Preflash	Before paper development	Additive raw exposure floor.
Print shape	Paper density response	Toe, straight-line, shoulder reshaping.
Bleach bypass	Film or print density	Retained silver / neutral density model.
Halation	Before film development	Back-reflection and scatter kernels.
Diffusion	Camera or print optics	Energy redistribution across blur scales.
Scanner	Post material density	Spectral scan, glare, MTF, unsharp masking.
HDR	Final display output	Reference white, peak luminance, transfer function.

7.25 Final Technical Advice

When the image looks wrong, identify the wrong stage before moving controls. A green cast from print timing, a wrong input transform, and a camera-side spectral filter can all look like “green,” but they mean different things and should be fixed differently.

The purpose of spektrafilm is to expressly not hide that complexity. It exposes enough of the process that a colorist can make stage-aware decisions.